

Energy Conversion Effect: A Universal Law of Matter and Energy in the Cosmic System

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Abstract

The traditional cognitive framework of "energy consumption" has long constrained the human understanding of the intrinsic relationship between matter and energy. Based on the Unified Theory of Cosmic Minimum Matter Units and the logical deduction of the cognitive boundary of human observation, this paper proposes the Energy Conversion Effect (ECE)—a universal basic law that runs through all material forms in the cosmic system. The core conclusion is that there is no absolute "energy consumption" in the cosmic material system; all material energy activities are essentially the directional conversion of one or more energy forms into other energy forms, accompanied by the adaptive evolution of material structure and the orderly transmission of energy information. This paper clarifies the scientific connotation, core characteristics and universal manifestation of ECE, expounds its internal connection with the basic laws of thermodynamics, and verifies its universality from the microcosmic particle level, the cosmic material level and the life body level. Furthermore, the paper reveals the application value of ECE in the research of life science and modern medicine, and provides a brand-new physical theoretical basis for the understanding of life energy metabolism, the essence of disease and the direction of medical regulation. ECE not only makes up for the cognitive deficiency of the traditional "energy consumption theory" in the interpretation of material-energy interaction, but also constructs a unified material-energy operation framework for the cosmic system, which is an important theoretical supplement to the Unified Theory of Cosmic Minimum Matter Units, and has important guiding significance for breaking the cognitive boundary of human beings and promoting the cross-development of physics, cosmology and life science.

Keywords: Energy Conversion Effect; ECE; Cosmic Material System; Matter-Energy Interaction; Energy Consumption; Life Energy Metabolism; Universal Physical Law

1 Introduction

For a long time, human beings have used the concept of "energy consumption" to describe the energy changes in material activities—from the operation of celestial bodies in the cosmic system to the metabolic activities of life bodies, from the

movement of microcosmic particles to the operation of artificial machines, all are defined as the process of "material consuming energy to maintain activities". This cognitive framework is based on the macroscopic empirical observation of human beings, and has played a role in the preliminary understanding of material-energy relationship in the early stage of scientific development. However, with the deepening of human research on the cosmic material system and the continuous breaking of cognitive boundaries, the limitations of the "energy consumption theory" have gradually emerged: it cannot explain the essence of energy conservation and transformation in the cosmic system at the fundamental level, cannot reveal the intrinsic connection between material structure evolution and energy form change, and even leads to the cognitive deviation of human beings on the energy operation law of life bodies in the fields of life science and medicine.

Based on the Unified Theory of Cosmic Minimum Matter Units—the core view that all cosmic materials are composed of the same basic minimum matter units and follow the unified operation law, combined with the logical deduction that "human observation tools cannot surpass the material basis of their own composition and thus cannot realize the ultimate observation of cosmic minimum matter units", this paper carries out an in-depth rethinking of the material-energy interaction relationship. Through the cross-dimensional deduction of microcosmic particle physics, cosmic cosmology and life system science, the paper proposes the Energy Conversion Effect (ECE), and clarifies its universality as a basic physical law.

The proposal of ECE is not a negation of the existing basic physical laws such as the law of conservation of energy, but a deepening and refinement of the cognitive level of energy conservation and transformation. It answers the core questions such as "what is the essence of material energy activity", "where does the 'consumed' energy go" and "what is the connection between energy change and material structure evolution" at the fundamental level of the cosmic material system. At the same time, ECE constructs a unified material-energy operation framework for the cosmic system, which provides a new theoretical perspective for the research of life science and modern medicine, and is of great significance for breaking the cognitive boundary of human beings on "energy consumption" and promoting the cross-integration of physics, cosmology and life science.

2 The Scientific Connotation and Core Characteristics of Energy Conversion Effect (ECE)

2.1 Scientific Connotation of ECE

Based on the Unified Theory of Cosmic Minimum Matter Units, the Energy Conversion Effect is defined as: in the closed cosmic material system, all material forms (including microcosmic particles, cosmic celestial bodies, life bodies, etc.) follow the law of conservation of energy in the process of energy activity; the energy involved in the

material activity does not disappear in the form of "consumption", but is directionally converted into one or more other energy forms according to the inherent structure and operation law of the material. This process is accompanied by the adaptive adjustment of the material structure, the orderly transmission of energy information and the maintenance of the dynamic balance of the material system, which is a universal and inevitable material-energy interaction phenomenon in the cosmic system.

The core connotation of ECE can be summarized as three points:

1. Non-consumption of energy: The essence of material energy activity is not "consumption" but "conversion"; the total energy in the cosmic material system is constant, and the energy involved in material activity only changes in form, not in absolute loss.
2. Directionality of conversion: The conversion of energy forms is not random, but is determined by the inherent structure of the material and the external environmental conditions; the material structure determines the direction and efficiency of energy conversion, and the energy conversion in turn promotes the adaptive evolution of the material structure.
3. Unity of material and energy: Energy conversion and material structure evolution are an organic whole; there is no energy conversion separated from material, and no material evolution without energy conversion; the two interact and restrict each other to maintain the dynamic balance of the material system.

2.2 Core Characteristics of ECE

As a universal basic law of the cosmic material system, ECE has four core characteristics that run through all material forms:

2.2.1 Universality

ECE is applicable to all material forms in the cosmic system, without the limitation of scale, structure and attribute—from the high-speed movement of microcosmic elementary particles (the conversion of kinetic energy into electromagnetic energy), to the orbital operation of cosmic celestial bodies (the conversion of gravitational potential energy into kinetic energy); from the photosynthesis of plants (the conversion of solar energy into chemical energy), to the metabolic activities of human bodies (the conversion of chemical energy into biological electrical energy, mechanical energy and thermal energy); all material energy activities follow the ECE, which is the inherent attribute of cosmic material.

2.2.2 Synergy

ECE is accompanied by the adaptive change of material structure and the orderly

transmission of energy information in the process of energy conversion; the three processes of energy form conversion, material structure evolution and energy information transmission are synergistic and inseparable. For example, the energy conversion of life body cells is accompanied by the change of cell membrane structure and the transmission of cell signal information; the energy conversion of stars is accompanied by the evolution of stellar structure and the transmission of cosmic electromagnetic information.

2.2.3 Dynamic Balance

The energy conversion process of material follows the principle of dynamic balance; the material system will adjust the direction and efficiency of energy conversion according to the internal and external environmental changes, so as to maintain the relative stability of the material structure and the normal operation of the system. When the dynamic balance of energy conversion is broken, the material structure will appear abnormal evolution, and even the material system will collapse. For example, the abnormal energy conversion of human body cells will lead to the disorder of cell structure and the occurrence of diseases; the abnormal energy conversion of stars will lead to the evolution of stellar structure and even the explosion of supernovae.

2.2.4 Irreversibility in a specific scope

In a specific time and space scope and material system, the energy conversion process of ECE has irreversibility—one or more energy forms can only be converted into specific energy forms under specific conditions, and cannot be reversed without external energy input. This characteristic is consistent with the second law of thermodynamics, and further explains the physical essence of entropy change in the material system: the entropy increase of the material system is essentially the result of the directional conversion of energy forms in the process of ECE, and the entropy change is the external manifestation of the dynamic balance of energy conversion.

2.3 The Internal Connection Between ECE and the Basic Laws of Thermodynamics

The proposal of ECE is based on the existing basic laws of thermodynamics, and is a deepening and refinement of the cognitive level of the law of conservation of energy and the second law of thermodynamics; the two are not contradictory, but a unified whole of "general law" and "specific interpretation".

1. ECE is the concrete embodiment of the law of conservation of energy at the material-energy interaction level: The law of conservation of energy clarifies that "energy cannot be created or destroyed, only converted from one form to another" at the macro level; ECE further reveals the intrinsic mechanism and universal law of energy conversion at the fundamental level of the cosmic material system, and answers the core questions such as "how energy is converted" and "what is the

connection between energy conversion and material structure", making the law of conservation of energy more concrete and detailed in the interpretation of material-energy interaction.

2. ECE explains the physical essence of entropy change in the second law of thermodynamics: The second law of thermodynamics points out that the entropy of an isolated system tends to increase, and the energy conversion has directionality; ECE further clarifies that the entropy increase of the material system is not the result of "energy consumption", but the result of the directional conversion of energy forms and the adaptive evolution of material structure in the process of energy conversion. The irreversibility of energy conversion in a specific scope is the intrinsic cause of entropy increase, and the dynamic balance of ECE is the fundamental guarantee for the relative stability of the material system entropy.

In short, ECE is not a subversion of the basic laws of thermodynamics, but a theoretical supplement and cognitive upgrade based on the existing laws; it makes the human understanding of the material-energy interaction law of the cosmic system rise from the "macroscopic law level" to the "fundamental mechanism level".

3 Universal Manifestation of ECE in the Cosmic Material System

ECE is a universal basic law of the cosmic material system, and its manifestation runs through all material forms and scale levels of the cosmic system. This paper verifies the universality of ECE from three core levels: microcosmic particle level, cosmic material level and life body level—the three levels are interrelated and mutually restricted, forming a complete cosmic material-energy conversion system.

3.1 The Manifestation of ECE at the Microcosmic Particle Level

The microcosmic particle is the basic material unit of the cosmic system, and its energy activity is the most primitive manifestation of ECE. At the microcosmic level, the energy activity of elementary particles (such as electrons, protons, photons) is essentially the directional conversion of energy forms, without any absolute "energy consumption".

For example, the high-speed movement of electrons in the atom is accompanied by the conversion of kinetic energy into electromagnetic energy; the transition of electrons from high energy level to low energy level releases photons, which is the conversion of electronic potential energy into photon energy (electromagnetic energy); the collision of elementary particles is the mutual conversion of kinetic energy, potential energy and mass energy (in accordance with Einstein's mass-energy equation $E=mc^2$). All these microcosmic particle energy activities follow the ECE: the energy of particles only changes in form, and the total energy is conserved; the energy conversion is accompanied by the change of particle motion state and the transmission of microcosmic information, and the direction and efficiency of

conversion are determined by the intrinsic attributes of particles and the external electromagnetic field conditions.

The manifestation of ECE at the microcosmic particle level is the fundamental basis of the universal existence of ECE; the energy conversion of all macroscopic materials is based on the energy conversion of microcosmic particles, and the microcosmic particle energy conversion system constitutes the material-energy foundation of the cosmic system.

3.2 The Manifestation of ECE at the Cosmic Material Level

At the cosmic material level (macroscopic level), the energy activities of cosmic celestial bodies (such as stars, planets, galaxies) are the macroscopic manifestation of ECE; the huge energy involved in the operation of cosmic celestial bodies is not "consumed" in the process of operation, but is continuously converted between different energy forms, maintaining the dynamic balance of the cosmic system.

For example, the nuclear fusion reaction inside the star is the conversion of mass energy into thermal energy, electromagnetic energy and kinetic energy; the thermal energy and electromagnetic energy released by the star are transmitted to the surrounding cosmic space, and are absorbed by other celestial bodies to participate in their energy conversion process; the orbital operation of planets around the star is the continuous conversion between gravitational potential energy and kinetic energy, and the slight energy loss of the planetary system (such as gravitational wave radiation) is essentially the conversion of gravitational energy into electromagnetic energy, which is still in line with the ECE.

Another example is the formation and evolution of galaxies: the gravitational contraction of interstellar gas and dust is the conversion of gravitational potential energy into thermal energy and kinetic energy; the collision and merger of galaxies are the mutual conversion of gravitational energy, kinetic energy and thermal energy. All cosmic material energy activities show the core characteristics of ECE: non-consumption of energy, directionality of conversion, unity of material and energy, and dynamic balance. The ECE at the cosmic material level is the macroscopic extension of the microcosmic particle ECE, and the two jointly maintain the stability and evolution of the cosmic material system.

3.3 The Manifestation of ECE at the Life Body Level

Life body is the advanced form of cosmic material evolution, and its energy metabolism is the most complex and delicate manifestation of ECE in the cosmic system; the life activity of all living organisms (from microorganisms to higher animals and plants) is essentially a continuous and orderly energy conversion process, and the so-called "energy consumption" in traditional life science is the cognitive deviation

caused by the macroscopic observation of human beings.

The energy metabolism of life body follows the ECE in the whole process: the energy obtained by the life body from the external environment (such as solar energy for plants, chemical energy for animals) is not consumed in the process of life activity, but is directionally converted into various energy forms needed for the life body's survival and development, accompanied by the synthesis and decomposition of biological macromolecules, the transmission of life information and the maintenance of the dynamic balance of the life system.

Taking the human body—the most complex life body in the known cosmic system—as an example, the core manifestation of ECE in human life activities is as follows:

1. The conversion of chemical energy into multiple energy forms: The human body obtains chemical energy from food (carbohydrates, proteins, fats, etc.); this chemical energy is not "consumed" in the process of human metabolism, but is converted into biological electrical energy (for nerve signal transmission and cell activity regulation), mechanical energy (for muscle movement, organ operation), thermal energy (for maintaining body temperature) and information energy (for cell signal transmission and genetic information expression).
2. The synergy of energy conversion and life structure evolution: The energy conversion of the human body is accompanied by the adaptive change of human tissue and cell structure; for example, the energy conversion of muscle cells is accompanied by the thickening of muscle fibers, and the energy conversion of nerve cells is accompanied by the extension of nerve synapses; the energy conversion promotes the evolution of human life structure, and the life structure in turn determines the direction and efficiency of energy conversion.
3. The dynamic balance of energy conversion and the maintenance of life activity: The human body adjusts the direction and efficiency of energy conversion in real time according to the internal and external environmental changes (such as exercise, rest, diet); when the energy conversion is in dynamic balance, the human body maintains a healthy state; when the dynamic balance of energy conversion is broken (such as the obstruction of energy conversion pathway, the reduction of conversion efficiency), the human body will appear sub-health or disease state.

The manifestation of ECE at the life body level is the highest form of the cosmic material ECE; it fully reflects the core characteristics of ECE such as universality, synergy, dynamic balance and irreversibility in a specific scope, and is the most direct basis for the application of ECE in life science and modern medicine.

4 The Application Value of ECE in Life Science and Modern Medicine

Life body is the advanced form of cosmic material evolution, and the life energy metabolism follows the ECE; this is the fundamental physical basis for the application

of ECE in life science and modern medicine. The proposal of ECE breaks the cognitive deviation of the traditional "energy consumption theory" in the research of life energy metabolism, and provides a brand-new theoretical perspective for the understanding of the essence of life activity, the essence of disease and the direction of medical regulation. It is an important theoretical bridge connecting cosmic physics and life science, and has important guiding significance for the development of modern medicine, especially integrative medicine and system medicine.

4.1 ECE Reconstructs the Cognitive Framework of Life Energy Metabolism

The traditional life science takes the "energy consumption theory" as the core, and regards the life energy metabolism as the process of "the life body consumes chemical energy from food to maintain life activity"; this cognitive framework only stays at the macroscopic empirical observation level, and cannot reveal the intrinsic mechanism of life energy metabolism.

Based on ECE, the cognitive framework of life energy metabolism is reconstructed: the life energy metabolism is not a "consumption" process, but an orderly and directional energy conversion process following the ECE; the life body is a high-precision energy conversion system of the cosmic material system, and the core of life activity is to maintain the dynamic balance of energy conversion, the smoothness of conversion pathway and the high efficiency of conversion efficiency.

This cognitive framework upgrade has three core significances for life science:

1. Revealing the intrinsic mechanism of life energy metabolism: It answers the core questions such as "what is the essence of life energy activity" and "where does the 'consumed' energy go" at the fundamental level, and makes the human understanding of life energy metabolism rise from the "empirical observation level" to the "theoretical mechanism level".
2. Constructing a unified energy metabolism framework for all living organisms: All living organisms (from microorganisms to higher animals and plants) follow the ECE in the process of energy metabolism; the difference of life energy metabolism is only the difference of energy conversion direction, efficiency and pathway determined by the life structure, and the essence is the same—this constructs a unified energy metabolism framework for all living organisms in the cosmic system.
3. Providing a new research direction for life science: The research of life science can shift from the traditional "how to supplement energy for life bodies" to "how to maintain the dynamic balance of life body energy conversion, unblock the energy conversion pathway and improve the energy conversion efficiency", which opens up a new research direction for life science.

4.2 ECE Reveals the Physical Essence of Disease from the Perspective of Energy Conversion

The essence of disease is the core scientific question of modern medicine; the traditional medicine (especially modern precision medicine) mostly attributes the disease to the abnormal structure of local tissues and cells (such as gene mutation, cell proliferation, organ damage), and ignores the intrinsic connection between local structural abnormality and the overall energy metabolism of the life body.

Based on ECE and the Life Conduction System Theory proposed in the previous research, this paper reveals the physical essence of disease from the perspective of energy conversion: the occurrence and development of all diseases are essentially the result of the broken dynamic balance of the life body energy conversion system in the process of ECE; the local tissue and cell structural abnormality (lesion) is the external manifestation of the abnormal life body energy conversion, and the core cause of disease is the obstruction of the energy conversion pathway, the reduction of conversion efficiency and the imbalance of conversion form of the life body.

The physical essence of disease based on ECE can be summarized as two core points:

1. Abnormal energy conversion is the root cause of disease: The life body maintains a healthy state by the dynamic balance of energy conversion; when the energy conversion pathway is blocked (such as the obstruction of nerve signal transmission leading to the block of biological electrical energy conversion), the conversion efficiency is reduced (such as the decline of cell metabolism leading to the reduction of chemical energy conversion efficiency) or the conversion form is unbalanced (such as the excessive conversion of chemical energy into thermal energy leading to the disorder of body temperature regulation) due to internal factors (such as genetic factors, aging) or external factors (such as diet, work and rest, emotion, environment), the dynamic balance of the life body energy conversion system will be broken, and the life body will appear sub-health or disease state.

2. Local lesion is the external manifestation of abnormal energy conversion: The abnormal energy conversion of the life body will first affect the "weak link" of the life structure (such as genetically susceptible organs, overloaded tissues and cells); the local tissue and cells will appear structural abnormality and functional disorder due to abnormal energy conversion, and then form the macroscopic lesion that can be detected by medical means. The local lesion is the "result" of abnormal energy conversion, not the "cause" of disease; this is the fundamental reason why modern precision medicine only focuses on local lesion intervention and often falls into the predicament of "treating symptoms but not root causes".

Taking the most common chronic metabolic diseases (diabetes, hypertension, obesity) as an example, the physical essence based on ECE is: diabetes is the obstruction of the chemical energy of food converted into the energy available for cells due to insulin resistance, leading to the imbalance of energy conversion form; hypertension is the obstruction of the biological electrical energy conversion of vascular smooth muscle

cells, leading to the abnormal contraction of blood vessels; obesity is the excessive conversion of chemical energy into fat storage energy due to the imbalance of energy conversion direction, leading to the accumulation of fat. All these chronic diseases are the result of the broken dynamic balance of the human body energy conversion system, and the local lesion (such as islet cell damage, vascular endothelial cell abnormality, fat cell proliferation) is the external manifestation of abnormal energy conversion.

4.3 ECE Defines the Core Direction of Modern Medical Regulation

The core direction of medical regulation is determined by the understanding of the essence of disease; the traditional modern precision medicine takes "eliminating local lesions and controlling abnormal indicators" as the core direction of regulation due to the cognitive deviation of the essence of disease, which leads to the clinical predicament of "treating symptoms but not root causes, high recurrence rate and many complications" in the treatment of chronic diseases, functional diseases and psychosomatic diseases.

Based on ECE and the physical essence of disease revealed in this paper, the core direction of modern medical regulation is redefined: the core of medical regulation is not to simply "eliminate local lesions" or "control abnormal indicators", but to take the restoration of the dynamic balance of the life body energy conversion system as the core goal, unblock the energy conversion pathway of the life body, improve the energy conversion efficiency and adjust the energy conversion form, so as to eliminate the root cause of disease and maintain the healthy state of the life body.

This core direction of medical regulation is a dialectical integration of microscopic precise intervention and macroscopic systematic regulation, which is specifically reflected in two aspects:

1. Microscopic precise intervention: eliminate the external manifestation of abnormal energy conversion: For the local lesions formed by abnormal energy conversion (such as tumors, nodules, organ damage), the microscopic precise intervention means of modern precision medicine (such as targeted therapy, minimally invasive surgery, gene intervention) are used to quickly eliminate or control the local lesions, relieve the clinical symptoms of patients, and create conditions for the restoration of the dynamic balance of the life body energy conversion system. This is the "symptomatic treatment" in medical regulation, and its core value is to quickly control the acute state of the disease and protect the life safety of patients.
2. Macroscopic systematic regulation: restore the dynamic balance of energy conversion: Taking the restoration of the dynamic balance of the life body energy conversion system as the core goal, through the comprehensive regulation means of diet, work and rest, exercise, emotion and traditional Chinese medicine conditioning, unblock the energy conversion pathway of the life body (such as improving nerve

signal transmission and blood circulation), improve the energy conversion efficiency (such as regulating cell metabolism and enhancing immune function), and adjust the energy conversion form (such as balancing the conversion of chemical energy into various energy forms), so as to eliminate the root cause of disease and prevent the recurrence of disease. This is the "etiological treatment" in medical regulation, and its core value is to restore the healthy state of the life body from the root cause.

In short, the core direction of modern medical regulation based on ECE is "symptomatic treatment" combined with "etiological treatment", microscopic precise intervention combined with macroscopic systematic regulation; this direction abandons the one-sidedness of traditional precision medicine that only focuses on local lesions, and returns to the essence of life body as a complete energy conversion system, which is the inevitable direction for the development of modern medicine.

5 Conclusion and Prospect

5.1 Conclusion

Based on the Unified Theory of Cosmic Minimum Matter Units and the logical deduction of human cognitive boundary, this paper proposes the Energy Conversion Effect (ECE)—a universal basic law that runs through all material forms in the cosmic system. The core conclusion of the paper is that there is no absolute "energy consumption" in the cosmic material system; all material energy activities are essentially the directional conversion of energy forms, accompanied by the adaptive evolution of material structure and the orderly transmission of energy information. The paper clarifies the scientific connotation and core characteristics of ECE, expounds its internal connection with the basic laws of thermodynamics, and verifies its universality from the microcosmic particle level, cosmic material level and life body level. Furthermore, the paper reveals the application value of ECE in life science and modern medicine, reconstructs the cognitive framework of life energy metabolism, reveals the physical essence of disease from the perspective of energy conversion, and redefines the core direction of modern medical regulation.

The main research achievements of this paper are as follows:

1. Proposing a new universal physical law: ECE is a new universal basic law of the cosmic material system proposed on the basis of the existing basic physical laws; it makes the human understanding of the material-energy interaction law rise from the "macroscopic law level" to the "fundamental mechanism level", and is an important theoretical supplement to the Unified Theory of Cosmic Minimum Matter Units.
2. Breaking the traditional cognitive deviation: ECE subverts the traditional "energy consumption theory" cognitive framework, clarifies that the essence of material energy activity is conversion rather than consumption, and provides a new theoretical perspective for human beings to understand the cosmic material system and life body

energy metabolism.

3. Constructing a cross-disciplinary theoretical bridge: ECE constructs a unified material-energy operation framework for the cosmic system, and builds an important theoretical bridge connecting cosmic physics, cosmology and life science; it provides a physical theoretical basis for the cross-development of the three disciplines, and promotes the integration of physics and life science.

4. Providing a new theoretical basis for modern medicine: ECE combined with the Life Conduction System Theory reveals the physical essence of disease from the perspective of energy conversion, and redefines the core direction of modern medical regulation; it provides a brand-new physical theoretical basis for the development of integrative medicine and system medicine, and is of great significance for breaking the clinical predicament of modern precision medicine.

5.2 Prospect

The proposal of ECE is a new attempt to break the human cognitive boundary and explore the universal law of the cosmic material system; with the deepening of human research on the cosmic system and the continuous development of cross-disciplinary science, the research of ECE will be further deepened and expanded, and its application value will be further reflected in more fields. The future research direction of ECE is mainly focused on the following four aspects:

1. The quantitative research of ECE: On the basis of the qualitative research of ECE, carry out the quantitative research of ECE, construct the mathematical model of energy conversion direction and efficiency of different material forms, and realize the quantitative description and prediction of the material energy conversion process; this is the key to the further development and application of ECE.

2. The verification of ECE in the extreme cosmic environment: Carry out the research of ECE in the extreme cosmic environment (such as black holes, neutron stars, cosmic microwave background radiation), verify the universality of ECE in the extreme environment, and further improve the theoretical system of ECE; this is an important direction for the combination of ECE and cosmology.

3. The empirical research of ECE in life science: Carry out the empirical research of ECE in the life body (such as the quantitative detection of life body energy conversion, the research of energy conversion pathway regulation), verify the manifestation and mechanism of ECE in the life body energy metabolism, and provide empirical evidence for the application of ECE in life science and modern medicine; this is the key to the landing application of ECE in the life science field.

4. The cross-disciplinary application research of ECE: Expand the application research of ECE in more cross-disciplinary fields (such as artificial intelligence, new energy, aerospace), take ECE as the theoretical basis, explore the new technology and new method of material-energy conversion, and promote the development of human science and technology; this is the ultimate goal of ECE research.

In the final analysis, the exploration of the cosmic material system is a process of continuous breaking of human cognitive boundaries; the proposal of ECE is just a new starting point for human beings to understand the material-energy interaction law of the cosmic system. With the continuous deepening of human research and the continuous expansion of cognitive boundaries, more universal basic laws of the cosmic system will be discovered, and human beings will move forward step by step towards the ultimate truth of the cosmic system.

The essence of the cosmic system is the unity of material and energy; all material activities follow the universal law of energy conversion—this is the core conclusion of ECE, and also the eternal direction of human beings to explore the cosmic system.

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